

Weekly Report

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Project: UAV-Based Traffic Safety & Video Analytics

Week: 28 Mar 2026 – 4 Apr 2026

Submission Date: 4 Apr 2026

Detailed Summary of Work Done During Current Week

A. Annotation Pipeline Utilization & Scaling

- Applied the semi-automated annotation pipeline on larger portions of the dataset.
- Successfully used the inference → CSV → XML → CVAT workflow to generate and refine annotations.
- Reduced manual effort by leveraging pre-annotated bounding boxes and correcting them in CVAT instead of labeling from scratch.
- Validated that the pipeline is scalable for handling long UAV video sequences (~9000 frames).

B. Annotation Quality Improvement

- Focused on improving annotation consistency across frames.
- Corrected bounding box inaccuracies and removed false detections during CVAT refinement.
- Ensured better alignment of annotations with vehicle positions across consecutive frames.
- Maintained more consistent labeling structure for downstream trajectory extraction.

C. Vehicle Trajectory Generation

- Generated trajectories for a larger number of vehicles using centroid-based extraction.
- Extended trajectory computation across longer continuous frame sequences.
- Visualized vehicle paths over time to validate movement continuity.
- Observed improved trajectory formation compared to previous weeks.

D. Identity Consistency Analysis

- Continued analyzing challenges related to maintaining consistent vehicle IDs across frames.
- Identified scenarios causing identity switches, such as:
 - Vehicle overlap
 - Occlusion
 - Dense traffic regions
- Evaluated need for improved tracking logic to maintain identity continuity.

E. Pipeline Integration and Workflow Maturity

- Strengthened integration between:
 - Detection (YOLO inference)
 - Annotation (CSV → XML → CVAT)
 - Trajectory extraction
- Established a clearer end-to-end workflow for processing UAV traffic videos.
- Organized outputs (annotations, trajectories, visualizations) into structured directories for reproducibility.

Tasks Planned for Coming Week

A. Trajectory Consistency Improvement

- Improve vehicle ID consistency across frames.
- Reduce identity switching during trajectory generation.
- Explore simple tracking enhancements for stable trajectory formation.

B. Behavior Analysis Initialization

- Begin mapping trajectories relative to predefined zones.
- Identify vehicle entry/exit patterns within zones.
- Prepare initial logic for traffic behavior analysis.

C. Final Pipeline Optimization

- Refine end-to-end pipeline for smoother execution.
- Improve efficiency for processing larger video segments.